Garbage-Guard Final Defense Script

Spoken script for Garbage-Guard Final Defense v6, cut to fit a 10-minute presentation. Facts as checked against Garbage-Guard v68.docx.

Speaker order

Speaker	Slides	Part	Approx. time
King Perth G. Pepito	1 to 4	Opening, the researchers, background of the study, research gap	about 2:15
Jade C. Gunayan	5 to 6	Objectives, scope and limitations	about 1:15
Brian Jay Celeste	7	Design: how Garbage-Guard works	about 1:00
Rodel S. Banogbanog	8	The prototype	about 0:30
Brian Jay Celeste	9 to 10	Dashboard and mobile app, development and testing	about 1:30
Genesis Walter D. Reyes II	11 to 15	Results, conclusions, recommendations, closing	about 2:45
Total	15 slides		**about 9:15 (limit 10:00)**

Times assume a calm pace of about 130 words per minute and leave room for pauses and slide builds. Lines in grey italics are cues and are not read aloud. Lines in blue italics are the handoff to the next speaker. On slide 10, click the V-Model or press V to enlarge it, then click again to close it.

King Perth G. Pepito (Slides 1 to 4)

Slide 1: Title

Good morning to the panel, to our adviser, Dr. Jonel R. Macalisang, and to everyone here.

We present Garbage-Guard: Development of Computer Vision for Creeks Garbage Detection Using Convolutional Neural Network. In short, we built a camera unit that watches a creek, counts the floating garbage, and warns the barangay when it piles up.

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Slide 2: The researchers

I am King Perth G. Pepito. I handled the systems analysis, the project rules and the proofreading. Genesis Walter D. Reyes II trained the detection model. Brian Jay Celeste worked on the system architecture, the model training and the Android app. Jade C. Gunayan handled the testing and the statistical analysis. And Rodel S. Banogbanog built the unit.

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Slide 3: Garbage in urban creeks

This is the creek at Barangay 870, Pandacan, where we tested the system.

Waste dumped into drains reaches the creek, builds up where the water slows, narrows the channel, and ends in flooding and stagnant, foul water. Today a barangay finds out about a build-up only when someone walks along the bank and looks.

So our question was simple. Can a camera tell the barangay when a creek needs clearing?

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Slide 4: The research gap

The studies we reviewed each solve only part of this. Detection studies find floating waste but do not count it or send an alert. Water-level sensors cannot tell what is in the water. Waste monitors watch bins, not moving water. And surveys and satellites give readings only once in a while.

Garbage-Guard puts four pieces together: a detector trained on the creek itself, counting only inside a drawn area of water, one ID per item so a drifting bottle is counted once, and a threshold alert with a saved snapshot.

Here is Jade Gunayan with our objectives and scope.

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Jade C. Gunayan (Slides 5 to 6)

Slide 5: Objectives

Thank you, King.

Our general objective was to develop a computer vision system for creek garbage detection using a convolutional neural network and YOLO26.

First, design a YOLO26n model that detects bottles, bags and food containers, measured by precision, recall and mAP at 0.5 on a held-out test set.

Second, build a Raspberry Pi 5 prototype that detects inside an area the operator draws, tracks each item, and alerts on a touchscreen and an Android app when the count reaches 20, the default.

Third, evaluate the system with ISO/IEC 25010:2023 for functional suitability, performance efficiency and usability, through developer tests and an end-user survey.

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Slide 6: Scope and limitations

The dataset was split 70, 10 and 20, and the test set was frozen, so every model version was graded on the same pictures. Our precision target was 80 percent.

As for the limits: the camera sees only floating, visible waste, the results depend on camera placement and daylight, and the system only monitors and warns. It does not collect garbage or test water quality.

Brian Celeste will now show how the system works.

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Brian Jay Celeste (Slide 7)

Slide 7: How Garbage-Guard works

Thank you, Jade. Let us follow one video frame from left to right.

The camera sends the video. The system crops the drawn area of water, and YOLO26n, in the NCNN format that runs fastest on the Pi, detects garbage at 960 pixels. ByteTrack gives each item an ID, so it is not counted twice. The count is the median of the last 15 frames, so one bad frame cannot cause a false alarm.

When the count reaches the threshold, the alert goes to the touchscreen, the buzzer, the phone through Firebase, and the history. It is raised once per build-up and is raised again only after the count falls to 15.

Rodel Banogbanog built the unit. Here he is.

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Rodel S. Banogbanog (Slide 8)

Slide 8: The prototype

Thank you, Brian.

Everything sits in a metal junction box, 25 by 30 by 15 centimeters, with the touchscreen in the door and a fan on each side. Inside are the Raspberry Pi 5, the SSD, and a UPS board with four 21700 cells, so the unit keeps running through a power cut. Only the camera is outside, mounted on the fence above the creek.

Brian Celeste continues with the dashboard and our testing.

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Brian Jay Celeste (Slides 9 to 10)

Slide 9: Dashboard and mobile app

Thank you, Rodel.

On the left is the touchscreen during an alert. Each item has a live box with its class, ID and confidence, and the count bar reads 20 of 20. In the middle is the phone alert, with the count, the limit and the snapshot. Both screens share one record, over the local network or through Firebase. From the settings, the user can change the threshold, control the buzzer, or upload a new model.

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Slide 10: Development and testing

The diagram at the top right is our method, a V-Model with Scrum sprints. Requirements and system design came first, together with their test plans. Software design, development and unit testing ran in sprints. System testing and acceptance testing closed the cycle.

The timeline shows the model versions. Version 5 added our first creek photos, version 7 froze the 807-image test set, and version 10 added photos from our third session at the creek.

Unit tests ran every sprint. System tests ran 100 trials over ten test cases. And acceptance testing was the field deployment on November 6, 7 and 9.

Genesis Reyes will present the results.

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Genesis Walter D. Reyes II (Slides 11 to 15)

Slide 11: Detection model results

Thank you, Brian.

On the frozen test set, precision is 84.6 percent, above our 80 percent target. Recall is 67.2 percent, so the model misses some items but rarely invents one. The mAP at 0.5 is 0.744.

On 19 creek pictures no model was trained on, the score rose from 0.149 for version 7 to 0.404 for version 10. These pictures come from the same spot and day as version 10's training photos, so they show this creek, not a new site. Bags were the weakest class. Pictures from the site itself improved the model more than any public dataset.

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Slide 12: System and field results

All ten system test cases passed in 10 out of 10 trials. One frame takes 275 to 350 milliseconds, in the Excellent band. The unit ran 24 hours over six days with no crash, freeze or overheating. Phone alerts arrived in 0.1 to 2 seconds on the local network, and in 5 seconds or more through the cloud. The counter passed all 19 automated tests.

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Slide 13: Field accuracy and evaluation

At the creek, live classification was 74.7 percent over three days, and it rose each time we lowered the camera: 62.5 percent at about 13 feet, 73.5 at 9 feet, and 85.0 at 7 feet. Even at 7 feet, the count still missed about half of the items.

Under ISO/IEC 25010, classification is Fair. Alerting, latency, stability and usability are Excellent, with usability at 4.40 out of 5 from four end users.

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Slide 14: Conclusions and recommendations

Objectives 1 and 2 were met: 84.6 percent precision, and one Raspberry Pi 5 that detects, tracks, counts, alerts and syncs on battery. For Objective 3, the system is Excellent for speed, stability and usability, and Fair for field classification at 74.7 percent.

Because the count runs low, alerts come late rather than false, so the threshold should be set from a hand count at each site.

We recommend mounting the camera low, collecting more bag images, retraining from the unit's own snapshots, and testing at night and in the rainy season. The other four recommendations are on the slide.

In short, Garbage-Guard is a workable early-warning tool for barangays, supporting SDGs 6, 9 and 11.

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Slide 15: Thank you

Thank you to the panel and to our adviser. We are now ready for your questions.